

Answer **all** questions. Answers must be written within the answer boxes provided. Working may be continued below the lines, if necessary.

- In Lucy's music academy, eight students took their piano diploma examination and achieved scores out of 150. For her records, Lucy decided to record the average number of hours per week each student reported practising in the weeks prior to their examination. These results are summarized in the table below.

Average weekly practice time (h)	28	13	45	33	17	29	39	36
Diploma score (D)	115	82	120	116	79	101	110	121

- Find Pearson's product-moment correlation coefficient, r , for these data. [2]
- The relationship between the variables can be modelled by the regression equation $D = ah + b$. Write down the value of a and the value of b . [1]
- One of these eight students was disappointed with her result and wished she had practised more. Based on the given data, determine how her score could have been expected to alter had she practised an extra five hours per week. [2]

[illegible]

Full marks are not necessarily awarded for a correct answer with no working. Answers must be supported by working and/or explanations. Solutions found from a graphic display calculator should be supported by suitable working. For example, if graphs are used to find a solution, you should sketch these as part of your answer. Where an answer is incorrect, some marks may be given for a correct method, provided this is shown by written working. You are therefore advised to show all working.

Section A

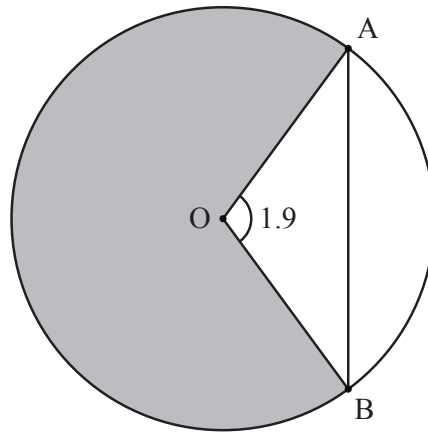
Answer **all** questions. Answers must be written within the answer boxes provided. Working may be continued below the lines, if necessary.

1. [Maximum mark: 6]

The following diagram shows a circle with centre O and radius 5 metres.

Points A and B lie on the circle and $\angle AOB = 1.9$ radians.

diagram not to scale



(a) Find the length of the chord $[AB]$. [3]

(b) Find the area of the shaded sector. [3]

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3. [Maximum mark: 5]

An arithmetic sequence has first term 60 and common difference -2.5 .

(a) Given that the k th term of the sequence is zero, find the value of k . [2]

Let S_n denote the sum of the first n terms of the sequence.

(b) Find the maximum value of S_n . [3]

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2. [Maximum mark: 6]

The value of a car is given by the function $C = 40\,000(0.91)^t$, where t is in years since 1 January 2023 and C is in USD(\$).

(a) Write down the annual rate of depreciation of the car. [1]

(b) Find the value of the car on 1 January 2028. [2]

Alvie wants to buy this car. On 1 January 2023, he invested \$15 000 in an account that earns 3% annual interest compounded yearly. He makes no further deposits to, or withdrawals from, the account.

Alvie wishes to buy this car for its value on 1 January 2028. In addition to the money in his account, he will need an extra \$ M .

(c) Find the value of M . [3]

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3. [Maximum mark: 6]

A company produces bags of sugar whose masses, in grams, can be modelled by a normal distribution with mean 1000 and standard deviation 3.5. A bag of sugar is rejected for sale if its mass is less than 995 grams.

- (a) Find the probability that a bag selected at random is rejected. [2]
- (b) Estimate the number of bags which will be rejected from a random sample of 100 bags. [1]
- (c) Given that a bag is not rejected, find the probability that it has a mass greater than 1005 grams. [3]

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4. [Maximum mark: 7]

A factory manufactures lamps. It is known that the probability that a lamp is found to be defective is 0.05. A random sample of 30 lamps is tested.

(a) Find the probability that there is at least one defective lamp in the sample. [3]

(b) Given that there is at least one defective lamp in the sample, find the probability that there are at most two defective lamps. [4]

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5. [Maximum mark: 7]

The population of a town t years after 1 January 2014 can be modelled by the function

$$P(t) = 15\,000e^{kt}, \text{ where } k < 0 \text{ and } t \geq 0.$$

It is known that between 1 January 2014 and 1 January 2022 the population decreased by 11 %.

Use this model to estimate the population of this town on 1 January 2041.

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5. [Maximum mark: 6]

The functions f and g are defined for $x \in \mathbb{R}$ by $f(x) = 6x^2 - 12x + 1$ and $g(x) = -x + c$, where $c \in \mathbb{R}$.

(a) Find the range of f . [2]

(b) Given that $(g \circ f)(x) \leq 0$ for all $x \in \mathbb{R}$, determine the set of possible values for c . [4]

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4. [Maximum mark: 8]

The weights, W grams, of bags of rice packaged in a factory can be modelled by a normal distribution with mean 204 grams and standard deviation 5 grams.

- (a) A bag of rice is selected at random.

Find the probability that it weighs more than 210 grams. [2]

According to this model, 80% of the bags of rice weigh between w grams and 210 grams.

- (b) Find the probability that a randomly selected bag of rice weighs less than w grams. [2]

- (c) Find the value of w . [2]

- (d) Ten bags of rice are selected at random.

Find the probability that exactly one of the bags weighs less than w grams. [2]

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Do **not** write solutions on this page.

8. [Maximum mark: 12]

A scientist conducted a nine-week experiment on two plants, A and B , of the same species. He wanted to determine the effect of using a new plant fertilizer. Plant A was given fertilizer regularly, while Plant B was not.

The scientist found that the height of Plant A , h_A cm, at time t weeks can be modelled by the function $h_A(t) = \sin(2t + 6) + 9t + 27$, where $0 \leq t \leq 9$.

The scientist found that the height of Plant B , h_B cm, at time t weeks can be modelled by the function $h_B(t) = 8t + 32$, where $0 \leq t \leq 9$.

- (a) Use the scientist's models to find the initial height of
 - (i) Plant B ;
 - (ii) Plant A correct to three significant figures. [3]
- (b) Find the values of t when $h_A(t) = h_B(t)$. [3]
- (c) For $0 \leq t \leq 9$, find the total amount of time when the rate of growth of Plant B was greater than the rate of growth of Plant A . [6]

